

Soft photonic skins with dynamic texture and colour control

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The visual appearances of surfaces are influenced by their colour and texture. Although the creation and tuning of structural colours has been realized with nanostructures^{1,2}, achieving dynamic control over visual texture^{3,4} remains challenging. Inspired by dynamic modulation of cephalopod skin^{5,6}, we develop polymer films with programmable surface textures. We bring these textures to life through immersion in different liquids that cause reversible local swelling/contraction to a degree that is determined by electron-beam irradiation. We show how standard electron-beam patterning tools can spatially encode arbitrary textures that can be hidden and shown on demand. Similarly, by modulating the topography of optical Fabry–Pérot cavities, we create colour patterns that can be continuously tuned with microfluidic control to achieve several distinct appearance states, allowing them to camouflage with different backgrounds. Finally, by creating multilayer devices, we demonstrate independent control of texture and colour in a single device, enabling a higher level of dynamic control over visual appearance.

Surface textures influence many aspects of our sensory perception of objects, ranging from tactile sensation to visual appearance. The latter is influenced by our perception of both the spectral and angular distributions of light waves. Living organisms frequently use the reversible deformation of soft tissue to modulate surface texture. This allows them to control interactions with their physical environment for camouflage and signalling. For example, actuation of muscle fibres in skin papillae allows cephalopods to rapidly transform their skin texture between a smooth, planar state and topographically complex, three-dimensional (3D) states that resemble natural textures found in their environment^{6–8} (Fig. 1a). These extreme ‘flat-to-3D’ textural changes, combined with the ability to change their skin’s colouration patterns through actuation of chromatophores, endow cephalopods with sophisticated control of their visual appearance⁵.

Many optical studies have achieved control over the spectral component of visual appearance, that is, colour, by engineering nanoscale resonances and wave-interference effects^{1,2,9–12}. However, the angular distribution of light is at least as important in determining visual appearance. Surface topography shapes this aspect of appearance by redistributing incident light waves based on the geometry and characteristic length scales of different surface features. This leads to the perception of properties ranging from visual texture, for example, whether the surface appears bumpy or wavy, to surface finish, for example, the degree of apparent gloss or haze. Dynamic control over surface textures, which are central in the way we perceive and interact with our environment^{3,12–14}, has remained elusive. Furthermore, independent control of texture and colour, which is key to the camouflage abilities of cephalopods, has yet to be realized.

Mutable soft materials can be used to dynamically vary surface topography^{15,16}. Dynamic modulation of texture using soft materials at both the microscale and the nanoscale could enable on-demand control over the scattering of light waves, providing new mechanisms for regulating visual appearance. However, present approaches to switchable surfaces, including recent demonstrations relying on elastic instabilities^{17–20} or patterning of hydrogels^{20–24}, are limited in terms of their degree of topographical contrast, their patternability or their achievable feature size. There is a clear need for methods capable of flexibly encoding a wide variety of lithographically defined, naturalistic textures that can be hidden and shown on demand.

Here we develop programmable surfaces based on thin polymer films in which the degree of swelling is spatially varied through dose-dependent electron-beam exposure. This can encode complex topographies across a broad range of length scales (millimetres to micrometres) that appear or disappear following immersion in different solvents (Fig. 1b). These flat-to-3D transformations, which switch between the extrema of topographic contrast, allow us to realize various textural and colourful appearances (Fig. 1c,d). We start by showing how textures structured at millimetre scales can be completely hidden and shown on demand. Second, by engineering switchable textures at the micrometre scale, we demonstrate switching between matte and mirror-like reflection, providing a tuning knob for under-explored properties of visual appearance such as gloss³. Third, by patterning soft-material-based optical cavities with complex topographies, we encode dynamic patterns of structural colour that can be tuned using microfluidics to access several distinct

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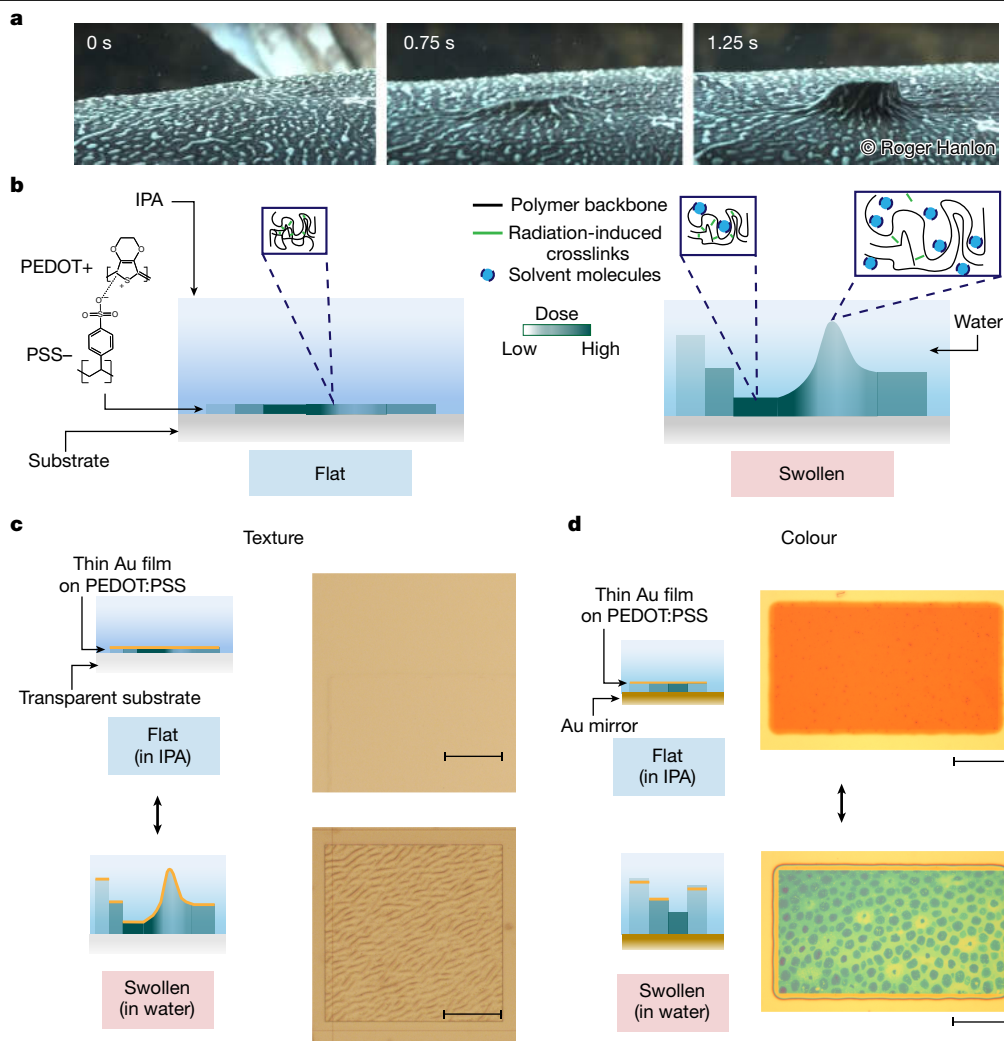


Fig. 1 | Cephalopod-inspired flat-to-3D transformations. **a**, Skin papillae expressed by cuttlefish, which extend or retract within about 1.3 s, provide exceptional camouflage capabilities. Frame grabs are from videos of a live animal, taken underwater in natural conditions in Australia. **b**, Schematic illustration of the working principle of polymer films with encoded flat-to-3D topographical transitions. The swelling degree of the polymer PEDOT:PSS is patterned by electron-beam exposure. In IPA, which is an unfavourable solvent, PEDOT:PSS does not swell, therefore the surface remains flat. In water, which is

a favourable solvent, PEDOT:PSS swells, transitioning to a complex surface topography encoded by the previously written dose pattern. **c**, Optical micrographs of patterned PEDOT:PSS films coated with a thin layer of metal. They show switching of visible texture between a flat state in IPA and a textured state in water. Scale bars, 40 μm . **d**, Optical micrographs of patterned Fabry–Pérot cavities, with an Au–PEDOT:PSS–Au structure. Their degree of visual contrast switches between a contrast-free state in IPA and a high-contrast state in water. Scale bars, 100 μm .

appearance states and camouflage with different backgrounds. Finally, by combining these elements in a multilayer device, we demonstrate independent, active control of both reflected colour and texture. This platform enables functional surfaces for applications ranging from dynamic camouflage and display for soft machines^{15,25} to new types of photonic device^{26,27}.

Achieving arbitrary control over surface topography

The realization of complex, reversible textures hinges on the ability to induce large changes in surface topographies across several spatial scales. We demonstrate that this can be achieved with the widely used conductive polymer, poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) and quantify the swelling of thin PEDOT:PSS films modulated by exposure to electron-beam radiation (Extended Data Fig. 1). We use this property to create programmable surfaces by first spatially patterning the degree of swelling of our surfaces and then dynamically modulating it by applying different solvents (Fig. 1 and Supplementary Fig. 1).

After electron-beam exposure, the degree to which the PEDOT swells in water is inversely proportional to the exposure dose (Supplementary Fig. 2). The volumetric swelling of isolated structures in water as compared with their dry state can be modulated from more than 800% at low doses down to 200% at higher doses (Fig. 2a and Supplementary Fig. 1). For a film with a thickness of 90 nm in the dry state, this difference leads to height variations approaching 550 nm between isolated regions patterned with different doses. The swelling is driven by uptake of water, which is a favourable solvent for the hydrophilic PSS component of PEDOT:PSS. Previous reports have demonstrated that PEDOT:PSS is a negative electron-beam resist^{28,29}, suggesting that electron-beam exposure reduces swelling through radiation-induced crosslinking³⁰ (Supplementary Text 1). The swelling degree achievable at a particular dose shows little dependence on the substrate material but is sensitive to the accelerating voltage (Supplementary Figs. 3 and 4).

The precise spatial control afforded by electron beams allows intricate and complex topographies to be produced in the swollen state. As an example, we patterned a nanoscale replica of the topography

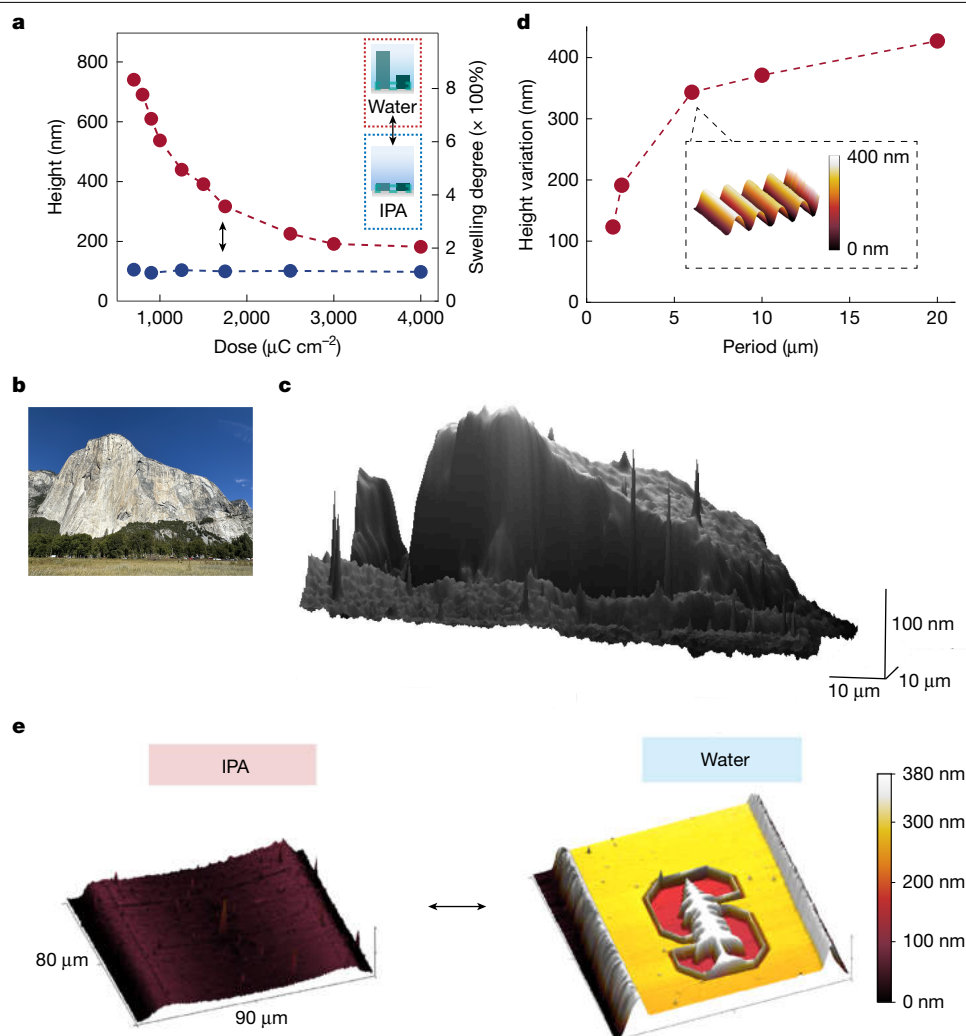


Fig. 2 | Spatial patterning of the swelling degree for control of complex surface topographies. **a**, Contrast curve for electron-beam-exposed PEDOT:PSS, which characterizes the swollen height in water (red) and IPA (blue) as a function of the electron dose for a film with a dry thickness of 90 nm. Measurements were made by sampling the edge height of isolated blocks in five locations. Data points shown are the mean; the standard deviation lies inside the plotted points. **b**, Photograph of El Capitan in California, USA. **c**, In situ liquid AFM of patterned PEDOT:PSS. A replica of El Capitan is encoded into the PEDOT:PSS in the swollen state. 3D data source: U.S. National Park

Service. Details of the dose pattern are included in Supplementary Fig. 4. **d**, The achievable height variation in the swollen state for a square-wave pattern of alternating high-dose ($2,500 \mu\text{C cm}^{-2}$) and low-dose ($500 \mu\text{C cm}^{-2}$) stripes at different periods applied to a film with a dry thickness of 90 nm. Measurements were made by sampling the height difference between the high-dose and low-dose regions in five locations. Data points shown are the mean; the standard deviation lies inside the plotted points. **e**, In situ liquid AFM of PEDOT:PSS encoded with a pattern of the Stanford seal, measured in the flat state in IPA and in the swollen state in water.

of the rock formation El Capitan in California (Fig. 2b,c). The 3D topographical data were discretized into 256 layers with different applied doses (Supplementary Fig. 5). Figure 2d shows the height variation for a continuous square-wave pattern of alternating high dose ($2,500 \mu\text{C cm}^{-2}$) and low dose ($500 \mu\text{C cm}^{-2}$) at different periods. As the period is reduced from $20 \mu\text{m}$ to $1.5 \mu\text{m}$, the maximum height variation is reduced from 427 nm to 123 nm . The achievable height variation for these continuously patterned structures depends on the dose variation (Supplementary Fig. 6) and is lower than the contrast achievable between isolated regions. This is because of electron-beam proximity effects and mechanical constraints on the swelling of adjacent structures (Extended Data Figs. 2 and 3). Increasing the dry film thickness to approximately 750 nm increases the maximum achievable height variation to more than $1.8 \mu\text{m}$ (Extended Data Fig. 4). Physical interactions at these microscopic length scales govern interfacial phenomena ranging from cellular adhesion and friction to reflection of light from surfaces. Structures patterned with dimensions comparable to the wavelengths of visible and infrared radiation could

shape optical wavefronts by controlling interference and diffractive effects.

Applying liquids with different solvation energies for PEDOT:PSS allows the flat state to be retrieved. Although the flat state can also be recovered by drying in air, controlling the topography while keeping the device in liquid allows for integration with fluidic control systems that are widely used in optofluidics and soft robotics^{25,30,31}. The height of isolated structures in 2-propanol (IPA), which does not swell PEDOT:PSS, is essentially the same as in the dry state and varies by less than 11 nm across all doses (Fig. 2a).

We demonstrate our ability to dynamically show virtually arbitrary, designer topographies by encoding different regions of the Stanford University seal pattern to demonstrate different degrees of swelling. The written structure transitions from a hidden, flat state in IPA to a structured state in water (Fig. 2e). This approach realizes a lithographically defined flat-to-3D transition representing the extrema of textural contrast, suggesting the potential to dynamically control properties of reflected light using fine, wavelength-scale structures.

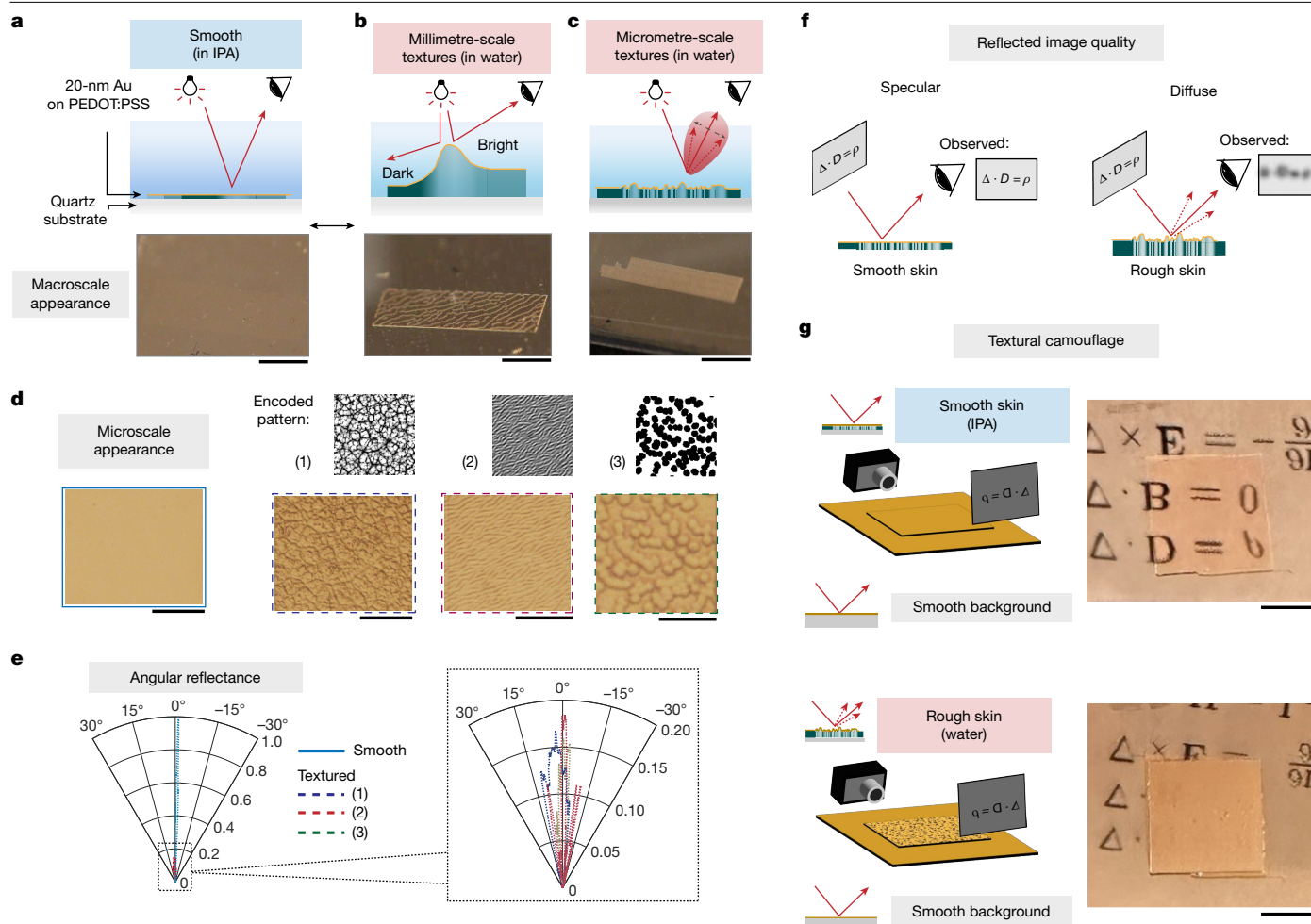


Fig. 3 | Dynamic control of surface texture at different length scales has distinct effects on visual appearance. **a–c**, Photographs illustrative of the macroscale appearance of textured surfaces. Scale bars, 2 mm. **a**, Textured surface with millimetre-scale features in its hidden, flat state in IPA. **b**, The same textured surface with millimetre-scale patterns swells in water to a state with visible wave-like textures. **c**, A separate textured surface with micrometre-scale features in its swollen state in water. **d**, Optical micrographs of surfaces with different microscale textures taken with a bright-field microscope under white-light illumination. Scale bars, 20 μ m. Optical micrographs (1)–(3) show the swollen state in water. A reference image for the flat state of (1) in IPA is depicted with a blue border. **e**, The angular reflectance spectra of the textures (1)–(3) from **d**, as measured by back-focal-plane imaging using a 600-nm

laser source. The colours in the plot correspond to the textures indicated by the colour of their surrounding box. **f**, Schematic illustrating the change in the angular distribution of reflected light from rough and smooth surfaces and the associated change in reflected image quality. **g**, Schematics of the experimental set-up and photographs illustrating dynamic changes in reflected image quality and textural matching. A textured surface (photonic skin), encoded with the pattern in **d** (1), was fabricated on a cover glass substrate. The photonic skin was placed on a gold-coated Si wafer, which served as a smooth reference background. Reflected images of printed text held near the skin and background surface were captured using a camera in the flat and swollen states. Scale bars, 2 mm.

Dynamic control of visible appearance

The perception of visual textures arises from the interaction of light with surface topographies. Surface features with different length scales control different visual phenomena. We first patterned the swelling of our thin films to create millimetre-scale visible surface textures by depositing a thin (approximately 20 nm) layer of metal on top of the polymer. Although the addition of metal reduced the swelling contrast (Supplementary Fig. 7), increasing reflectance allowed for easy observation of encoded textures with the naked eye. Figure 3a,b shows a ripple-like texture pattern that is invisible in the flat state (Fig. 3a) but becomes visible in the swollen state (Fig. 3b). Their swollen state surface topography, patterned at feature sizes on the order of hundreds of μ m, is gently sloped (Supplementary Fig. 8). These features, which are hundreds of times larger than the wavelengths of visible light, specularly reflect incident light in a variety of directions.

A region of the surface that redirects light rays in directions falling inside the collection solid angle of the eye (or other optical observation system) will appear brighter, whereas regions that primarily reflect rays outside the collection solid angle will appear darker. These alternating regions of bright and dark create the visual appearance of texture.

Patterning structures at the microscale with feature sizes of about 2–4 μ m affords control over the overall visual perception of the surfaces. In the swollen state, these microtextured surfaces have a rough, matte-like appearance (Fig. 3c). This is because of diffuse scattering caused by diffractive effects occurring at these patterning length scales. Figure 3d shows optical micrographs of a range of naturalistic textures written with microscale feature sizes. We investigate the angular distribution of reflected light through back-focal-plane imaging (Supplementary Fig. 9). In the flat case, the reflected angular spectrum is concentrated within 1° of the specular direction (Fig. 3e). In the swollen case the reflected light becomes substantially more diffuse. The peak

intensity in the specular direction is reduced by more than 80% and the light is redistributed to angles up to 10° around the specular direction (Fig. 3e (1)–(3)). The angular spectrum depends on the spatial frequency content of the texture patterns whereas the diffuse intensity depends on the height variation^{32,33}. This shows the possibility of engineering reflected light fields with desired angular distributions, such as uniform diffusers (Fig. 3e (1)) or grating-like structures that can redirect light into preferred directions (Fig. 3e (2)).

We illustrate how such microscale patterning provides control over the visual perception of surfaces by demonstrating dynamic modulation of the haze of a surface (Fig. 3f). Switching from a smooth to a hazy surface appearance degrades the quality of a reflected image containing printed text, with notable blurring of characters (Fig. 3g).

In contrast to previous reports, which control the angular distribution of light with resonant nanostructures^{3,13,14} or reflective inks^{4,34}, our approach allows for control of angular distribution that: (1) can be dynamically switched on and off and (2) modifies the distribution without correlated colour changes. This approach may provide a route to dynamic textural camouflage, in which photonic skins change their visual appearances to match the properties of different backgrounds (Extended Data Fig. 5).

Dynamic colour control with Fabry–Pérot resonances

In this section, we show how precise spatial control over swelling behaviour can generate complex, dynamically switchable colour patterns. To this end, we placed PEDOT:PSS between two thin metallic layers to create Fabry–Pérot resonators with a spectral reflectance that is controlled by the thickness of the polymer layer (Fig. 4a). In the flat state, the surfaces show almost no observable colour contrast, whereas in the swollen state, the topography translates into a visible colour pattern (Fig. 1d and Fig. 4b). The large variations in swelling degree that can be achieved allow for the creation of optical cavities with greatly varying mirror spacings and, thus, many colours (Extended Data Fig. 6). The accessible colour space can be expanded by using alternative metals such as aluminium that have different optical properties³⁵ (Extended Data Fig. 7).

On switching from IPA to water using a liquid flow cell (Supplementary Videos 1 and 2), the surface switches between a low-contrast, ‘hidden’ state to a high-contrast state showing distinct hues (Fig. 4c). Observing the switching dynamics reveals that 90% of the colour change occurs in less than 10 s. The full contrast change is achieved within 20 s during swelling and 25 s during de-swelling. Robust switching of visual contrast is maintained after 250 cycles (Extended Data Fig. 8 and Supplementary Text 2). Although active control of structural colour has been widely explored^{36–39}, it remains challenging to rapidly transition from a uniform contrast-free state to a high-contrast state. The ability to hide and show information on demand could lead to applications ranging from encryption⁴⁰ to camouflage^{15,25,41}.

Checkerboard patterns with pixels of several distinct colours allow the spatial resolution of the patterning process to be evaluated (Fig. 4d). The width of the interface between two pixels varies from about 0.8 µm to 1.5 µm depending on their dose difference, with higher dose and hence colour differences showing wider interfaces.

The fine analogue control over cavity thickness afforded by our method allows us to smoothly represent subtle gradations of colour (Fig. 4e). This results in rich images capable of conveying a sense of depth. We demonstrate this effect by translating the greyscale of a naturalistic image of seagrass (Fig. 4f) into a 256-level dose pattern that is used to define the topography of an optical cavity (Fig. 4g). Blades of grass in the optical micrograph of this cavity exhibit subtle variations in hue that communicate features such as curvature (Fig. 4h,i). Notably, the surface topography also affects the angular distribution of light by scattering (Supplementary Fig. 11a) and redirecting reflected light to different angles (Supplementary Fig. 11b). This ultimately lends

textural richness to images and allows texture to be perceived even without cavity-induced colour gradients (Supplementary Fig. 11c).

Cephalopods camouflage by controlling both the pattern contrast and colouration of their skin^{7,8}. We demonstrate that cavity-based photonic skins can show several distinct colour and contrast states by applying solvent mixtures with different ratios of IPA and water. This provides access to intermediate states between the swelling extrema, enabling camouflage against different backgrounds. Regions of high and low doses access different colour trajectories as the device swells, achieving many distinct combinations of colours. As a demonstration, we create a pattern consisting of dots exposed to higher doses superimposed on a uniform plane exposed to a low dose. This evolving dot/plane colour pattern achieves five distinct appearance states as it swells (Supplementary Fig. 12), with varying degrees of contrast. Precisely modulating the water/IPA ratio with microfluidics dynamically controls the appearance of the skin (Supplementary Videos 3–5). In this way, the appearance can be tuned to best blend in with different backgrounds (Fig. 4j–m). Given the intricacies of human visual perception, the device can achieve impressive camouflage even when its accessible colours and geometries only imperfectly match the background.

The broader uptake of our surfaces will be assisted by the industrial availability of PEDOT:PSS and its established integration in device-fabrication processes in fields ranging from bioelectronics to optoelectronics. We highlight the straightforward integration of our thin-film system into different form factors by fabricating a device on a flexible polyimide substrate (Extended Data Fig. 9). Flexible photonic skins may have applications ranging from wearable devices to coatings for soft machines.

Independent control of texture and colour

The extraordinary camouflage abilities of cephalopods rely on simultaneous and independent control of texture and colour patterns. By patterning optical cavities and textured surfaces on either side of a thin transparent substrate, we achieve independent control of distinct texture and colour patterns by exposing each side to a solvent of choice (Fig. 5a). Exposure to water on both sides generates a coloured, textured state (Fig. 5b). The observed appearance is defined by both the pattern of dark and bright regions set by the texture pattern as well as the colour pattern set by the optical cavities. Switching to IPA onto different sides of the substrates causes de-swelling of either the colour or the texture pattern, which enables three appearance states: (1) a flat state with no colour or texture pattern (Fig. 5c); (2) a texture-only state (Fig. 5d); and (3) a colour-only state (Fig. 5e). The integration of several partially transmissive layers allows us to create devices that provide a new level of active control of visual appearance.

Outlook

By providing a means to actively modulate texture and colour simultaneously, we enable new approaches to dynamic camouflage, along with other applications that may benefit from access to a myriad of complex visual appearance states, including encryption⁴⁰, dynamic artworks and displays. At present, we transition between a flat state and a single encoded geometric pattern, whereas cephalopods can achieve a diverse range of patterns⁶. Recent studies have shown that PEDOT:PSS swells/de-swells reversibly in response to changing electrochemical potentials^{42,43}. This suggests the possibility to electrically tune texture and colour patterns using addressable pixelated devices. Computer-vision algorithms could guide control systems for devices that camouflage in arbitrary, unknown environments. Multibeam electron-beam lithography systems that are at present under development in the semiconductor industry may enable manufacturing of small-scale devices⁴⁴. Developing alternative methods of patterning

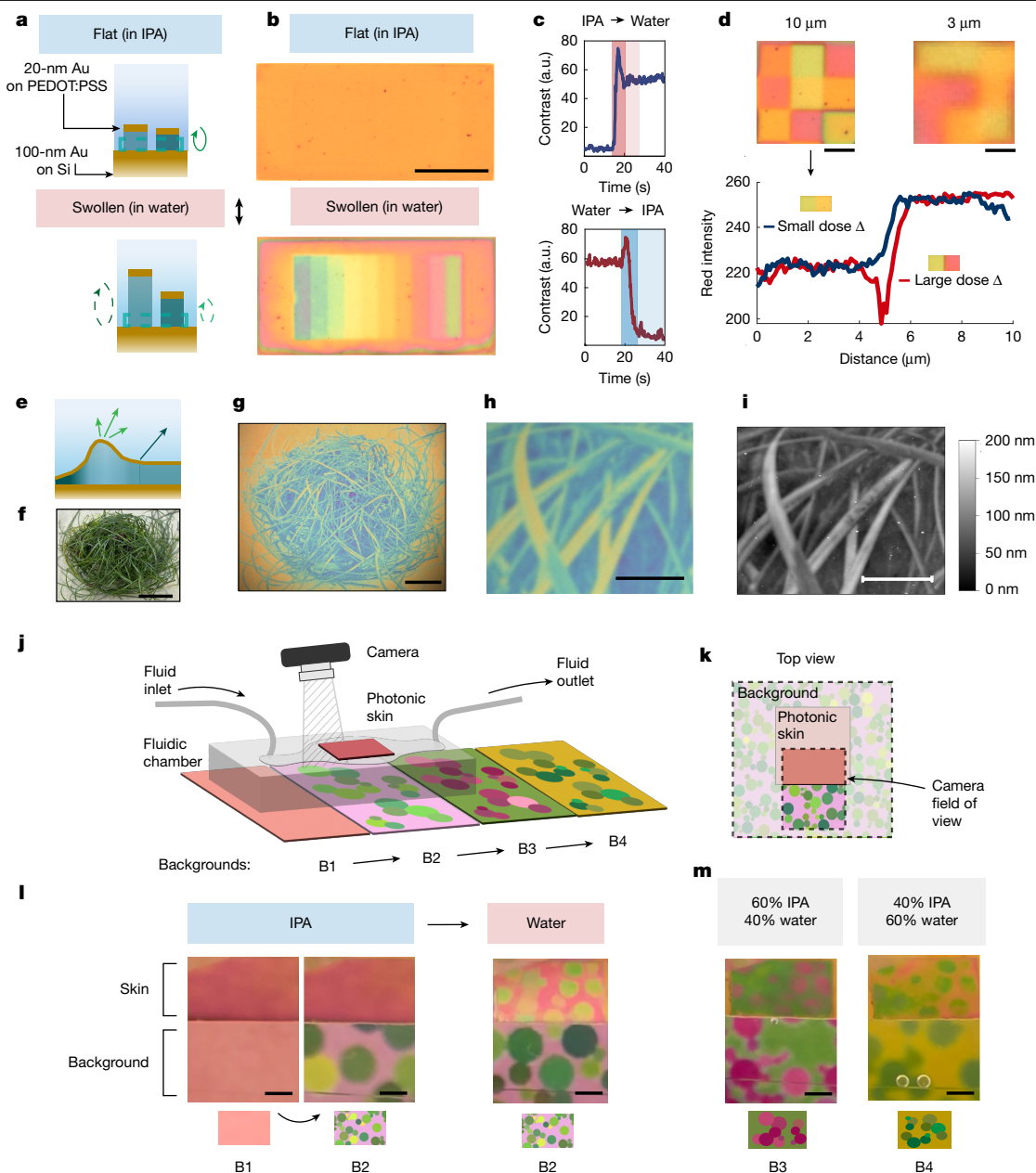


Fig. 4 | Dynamic control of colour patterns with topographically patterned optical cavities. **a**, Schematic illustrating differential swelling of optical cavities in different solvents. **b**, Micrographs of a cavity patterned with stripes of different electron-beam doses (see Methods) in IPA and water. Scale bar, 30 μm . **c**, Evolution of the colouration contrast, defined as the difference between the highest and lowest greyscale pixel intensity values, when switching the solvent from IPA to water (blue curve) and from water to IPA (red curve). Dark red and blue shadings indicate the period in which 90% of the contrast changes, whereas light red and blue bars indicate the period in which the remaining 10% of the contrast changes. a.u., arbitrary units. **d**, Micrographs of cavities arranged in patterns of square pixels, with pixel pitches (and scale bars) of 10 μm and 3 μm . The intensity of the red RGB channel was plotted across the interface of adjacent pixels exposed to different doses (see Methods). The specific pixel interfaces analysed are shown in the insets. **e**, Schematic of topographically complex cavities that reflect different colours and redirect light. **f**, Photograph of marine surfgrass. Scale bar, 10 cm. **g**, Optical micrograph

of a topographically complex cavity fabricated using a dose pattern based on the greyscale intensity of the image in Fig. 4f. Scale bar, 100 μm . **h**, **i**, In situ liquid AFM and optical micrograph of a section of the optical cavity shown in Fig. 4g. Scale bars, 30 μm . **j**, Schematic of the system used for background matching. A patterned optical cavity (photonic skin) was placed in a transparent microfluidic chamber above a slider holding a set of printed colour patterns. The target background pattern was changed by moving the slider. The appearance of the skin was changed by applying different ratios of IPA/water using a microfluidic controller (not shown) while being imaged from the top. **k**, Schematic showing the top view and the field of view observed by the camera. **l**, Images of the skin partly covering the background for various backgrounds and solvent ratios. Scale bars, 3 mm. The device begins in a uniform state in IPA. The background is changed from B1 to B2 by physically moving the slider. Water is applied to the device to adjust its state. **m**, Skin with the IPA/water ratios adjusted to provide the skin appearance that best blends in with backgrounds B3 and B4. Scale bars, 3 mm.

the swelling of PEDOT:PSS, including chemistries compatible with ultraviolet (UV) photolithography⁴⁵, or larger-area thermal methods⁴⁶ could enable bulk manufacturing. Finally, integrating our surfaces into macroscale soft robotic systems could enable truly cephalopod-like

camouflage¹⁵. These efforts will be aided by the established integration of fluidic systems for actuation and display in a wide range of technologies, from mature reflective displays^{47,48} to emerging wearable devices⁴⁹ and soft machines^{25,30}.

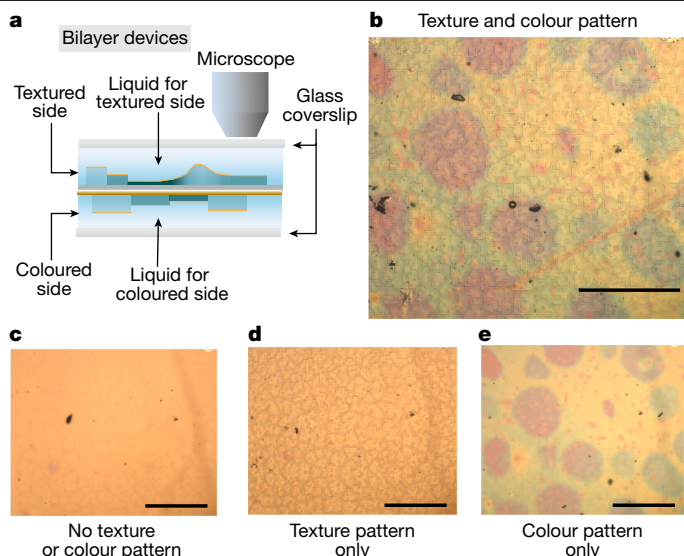


Fig. 5 | Dynamic and independent control over texture and colour with multilayer devices. **a**, Schematic illustration of a bilayer device structure and our microscope imaging configuration. The textured side consists of a 20-nm metal-coated, electron-beam-patterned PEDOT:PSS film fabricated directly on the coverslip. The coloured side consists of optical cavities formed from PEDOT:PSS films between two 20-nm-thick layers of Au that are patterned on the backside of the coverslip. Separate solvents can be independently applied to either side of the coverslip using a fluid cell. **b–e**, Optical micrographs of our device with: water on both sides of the device (**b**); IPA on both sides of the device (**c**); water on the texture side and IPA on the colour side (**d**); IPA on the texture side and water on the colour side (**e**). Scale bars, 200 μm .

Beyond the aesthetic possibilities, tunable photonic skins may enable new approaches to light control. The metal layer can be patterned to form arrays of nanoscale optical resonators, or optical metasurfaces, that scatter light with engineered spectral, angular and polarization distributions⁴³. The interplay between these optical resonances and complex geometry provides rich possibilities for creating unconventional appearances³ and dynamically shaping complex light fields^{26,27}.

Finally, microscale and nanoscale geometry defines boundary conditions that govern a range of physical phenomena, from friction and adhesion to fluid flow. Our programmable surfaces could enable new approaches to applications in which the microscale and nanoscale geometry dictates performance⁵⁰, including programming of cell growth⁵¹, control of friction⁵², reconfigurable microfluidics⁵³, electronics⁵⁴ and 3D biological interfaces⁵⁵.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09948-2>.

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Methods

Materials

Aqueous dispersions of the chemical PEDOT:PSS (Heraeus Clevios PH1000, 1.3 wt%) were used as received.

Sample fabrication: film preparation

Programmable surfaces were prepared by exposing thin films of spin-coated PEDOT:PSS to electron-beam radiation followed by development in water (Supplementary Fig. 1).

Before spin-coating, substrates were cleaned by sonication (5 min) in acetone and IPA, followed by 13 min of UV-ozone treatment to improve the wettability for spin-coating. Devices for process characterization and demonstrations of visible texture (Figs. 1c, 2a,b and 3) were prepared on bare quartz wafers. Devices demonstrating patterning of the Stanford logo and Yosemite topography were prepared on bare Si wafers. Devices demonstrating optical cavities (Fig. 4) were prepared on Au-coated Si substrates, prepared by electron-beam evaporation of Ti (3 nm) followed by Au (100 nm) onto a bare Si wafer. Multilayer devices (Fig. 5) were prepared using a 150- μm -thick glass coverslip (purchased from VWR, #1 size). In this instance, the metal cavity base layer was prepared by electron-beam evaporation of Ti (3 nm) followed by Au (100 nm).

PEDOT:PSS was then spin-coated at 2,000 rpm for 60 s for all measurements, with the exception of the camouflage experiments in Fig. 4j–m (which were spin-coated at 1,000 rpm for 60 s) and thick-film samples measured in Extended Data Fig. 4 (which are described in further detail below). Samples spin-coated at 2,000 rpm for 60 s have a dry film thickness of about 90 nm. Samples spin-coated at 1,000 rpm for 60 s have a dry film thickness of about 120 nm. To prepare thick film with a dry thickness of about 750 nm, PEDOT:PSS layers were spin-coated from a high-solid-content PEDOT:PSS dispersion that was obtained with rotary evaporation. 30 ml of PH1000 PEDOT:PSS in an open beaker was heated to 85 °C for 1 h while stirring with a magnetic bar at 350 rpm. For thicker films, 30 ml of PH1000 PEDOT:PSS was heated to 100 °C for 1.5 h while stirring. After 15 min of sonication and filtering with a 0.45- μm syringe, filter films were spin-coated at 1,000 rpm for 3 min. Samples were allowed to dry in an ambient environment without post-baking.

Sample fabrication: electron-beam exposure

Electron-beam patterning was performed by exposing the PEDOT:PSS films with either a Raith VOYAGER electron-beam lithography system using an acceleration voltage of 50 kV and an aperture of 60 μm or a Raith EBPG 5200+ system using an acceleration voltage of 100 kV. Samples in Fig. 4d,j–m, Extended Data Figs. 2, 4 and 7 and Supplementary Figs. 4, 10 and 12 were exposed at an acceleration voltage of 100 kV. All other samples were exposed at an acceleration voltage of 50 kV. The typical doses used for patterning 90-nm-thick samples with a 50-kV electron-beam lithography tool fall between 500 and 4,000 $\mu\text{C cm}^{-2}$ and between 1,000 and 8,000 $\mu\text{C cm}^{-2}$ with a 100-kV electron-beam lithography tool. Thicker films required higher minimum doses (Extended Data Fig. 4).

The doses used for samples shown in the figures in the main text are listed in Extended Data Table 1. Samples shown in the Extended Data Figs. and Supplementary Figs. have any relevant dose information listed in their captions. Per-pixel dose assignments for textures converted from greyscale images were performed using the 3D-PEC module of the BEAMER software (GenlSys GmbH) using the measured contrast curves for the relevant acceleration voltage (Fig. 2a and Supplementary Fig. 4).

After the electron-beam exposure, samples were developed in deionized water for 90 s, followed by drying with a nitrogen gun. Samples can also be dried with air.

Sample fabrication: metal evaporation on PEDOT:PSS

Thin metal films were subsequently deposited onto the exposed PEDOT:PSS surfaces by electron-beam evaporation. The metal top

layer in all instances was a 20-nm Au film, except for the sample used to evaluate reflected image quality in Fig. 3f, for which we used a 3-nm Au–7-nm Ag–3-nm Au trilayer film, and Extended Data Fig. 7, for which we used a 4-nm Au/11-nm Al film.

In situ liquid AFM measurements

Atomic force microscopy (AFM) images were collected in both water and IPA using an Asylum MFP-3D atomic force microscope in tapping mode. A fluidic cell was used to hold the liquid. Gold-coated, HQ:NSC35/Cr-Au BS cantilevers (MikroMasch) with a force constant of 5.4 N m⁻¹ were used. Images with maximum dimensions of 90 μm were collected with scan rates between 0.5 and 1.0 Hz, drive amplitudes of about 1 V and amplitude set points of 200–300 mV. Images were processed with the Gwyddion software.

Liquid flow cell for microscope-based optical measurements

We used a homebuilt liquid flow cell to allow for fluidic switching of polymer swelling under a microscope objective. Flow cells were constructed by sandwiching two pieces of pre-cut double-sided adhesive tape between the sample and a glass coverslip. The sample area was larger than the coverslip area. Solvent pipetted onto the sample near the periphery of the coverslip was then drawn into the flow cell by capillary action. This system allowed for fluidic switching while maintaining optical access for microscopy.

Reflectance measurements and spectral characterization

Reflectance measurements and spectral characterization were carried out in reflection mode using a Nikon C2 confocal microscope with a true-colour CCD (Nikon) to capture images. Samples were illuminated with an incident beam of unpolarized white light from a halogen lamp. They were viewed in bright field through a coverslip-corrected objective (Nikon numerical aperture (NA) 0.4, 20 $\times$) for all measurements except for Supplementary Fig. 11a,b,i. and ii., in which a higher-magnification dark-field-capable objective was used (Nikon NA 0.6, 50 $\times$). To acquire the reflectance spectra in Extended Data Fig. 6, we use a spectrometer (Acton 2300i, SpectraPro) with a cooled CCD (Princeton Instruments, Pixis 1024). The spectra are normalized by subtracting the CCD dark counts from the experimental spectrum, then the difference is divided by the spectrum of the halogen lamp collected with a Ag mirror used as a reflective reference standard (Thorlabs).

Fourier imaging for angular distribution measurements

For the angular distribution characterization, a quasi-monochromatic beam (bandwidth approximately 5 nm) is first generated by an NKT SuperK EXTREME supercontinuum laser with an acousto-optic tunable filter. The beam is unpolarized. Light was transmitted to the sample through an objective lens (Nikon NA 0.4, 20 $\times$ for all measurements except for Supplementary Fig. 11a,b, in which a higher-magnification NA 0.6, 50 $\times$ objective was used). A Bertrand lens was also used to directly image the Fourier plane (Fig. 3 and Supplementary Figs. 8b and 11b).

Fabrication of microfluidic devices by stereolithography 3D printing

All 3D-printed moulds used for this work were designed in Onshape CAD software, exported as an STL file and printed using a CADworks3D ProFluidics 285D printer (MiiCraft, CADworks3D) using the Master Mold for PDMS Photopolymer Resin (CADworks3D). After printing, the moulds were rinsed with isopropyl alcohol for 1 min, followed by a 1-min submersion in water to remove any uncrosslinked resin. Printed moulds were dried with compressed air and post-cured for 20 min on either side using the CureZone UV box (CADworks3D). Devices were made from the printed moulds by pouring RTV615 precursor solutions at a ratio of 10:1 (base:curing agent) (R.S. Hughes), degassed and cured for 15 min in an 80 °C oven. Once cured, inlet and outlet

Article

ports were created using a 0.75-mm biopsy punch (World Precision Instruments). PDMS and glass slides or coverslips were treated with air plasma for 45 s in a PDC-001 plasma cleaner (Harrick Plasma) brought into contact and placed in an 80 °C oven to covalently bond the pieces.

Operation of devices with a pressure-based microfluidic flow controller

All fluid flow through the microfluidic devices was driven using a pressure-driven microfluidic control system MFCS-EZ with OxyGEN software (Fluigent). Pressurized air from the MFCS-EZ drove flow of fluids from large reservoirs for continuous operation (Microfluidic Reservoir for 50 ml Falcon Tube M 4-port, Elveflow). Flow pressures did not exceed 500 mbar. All solutions were introduced to the devices by means of a blunt steel pin (O.D. 0.025 in, I.D. 0.013 in; New England Small Tube NE-1310-02) and Tygon ND 100-80 tubing (O.D. 0.06 in, I.D. 0.02 in; Fisher Scientific 14-171-284).

Data availability

Source data are provided with this paper. Raw AFM data files are available at the following Zenodo repository: <https://doi.org/10.5281/zenodo.17347008> (ref. 56).

56. Doshi, S. AFM data for “Soft photonic skins with dynamic texture and color control”. Zenodo <https://doi.org/10.5281/zenodo.17347008> (2025).

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Author contributions S.D. found that the dispersibility of PEDOT:PSS and its swelling in water is modulated by electron-beam exposure. S.D., M.L.B. and N.A.M. conceived the project. S.D., N.A.G., G.D., J.C. and B.L. performed sample fabrication. S.D. performed optical measurements. S.D. and J.C. performed AFM. S.D., J.D. and B.L. developed workflows for greyscale file preparation. J.E.O.-C. and P.S. designed and fabricated the microfluidic devices. S.D. performed the analysis. S.D., M.L.B. and N.A.M. designed the figures and wrote the manuscript, with assistance from N.A.G., G.D. and A.S. The research was supervised by M.L.B., N.A.M., A.S. and P.M.F.

Competing interests S.D., N.A.M. and M.L.B. have filed a patent application related to this work. The other authors declare no competing interests.

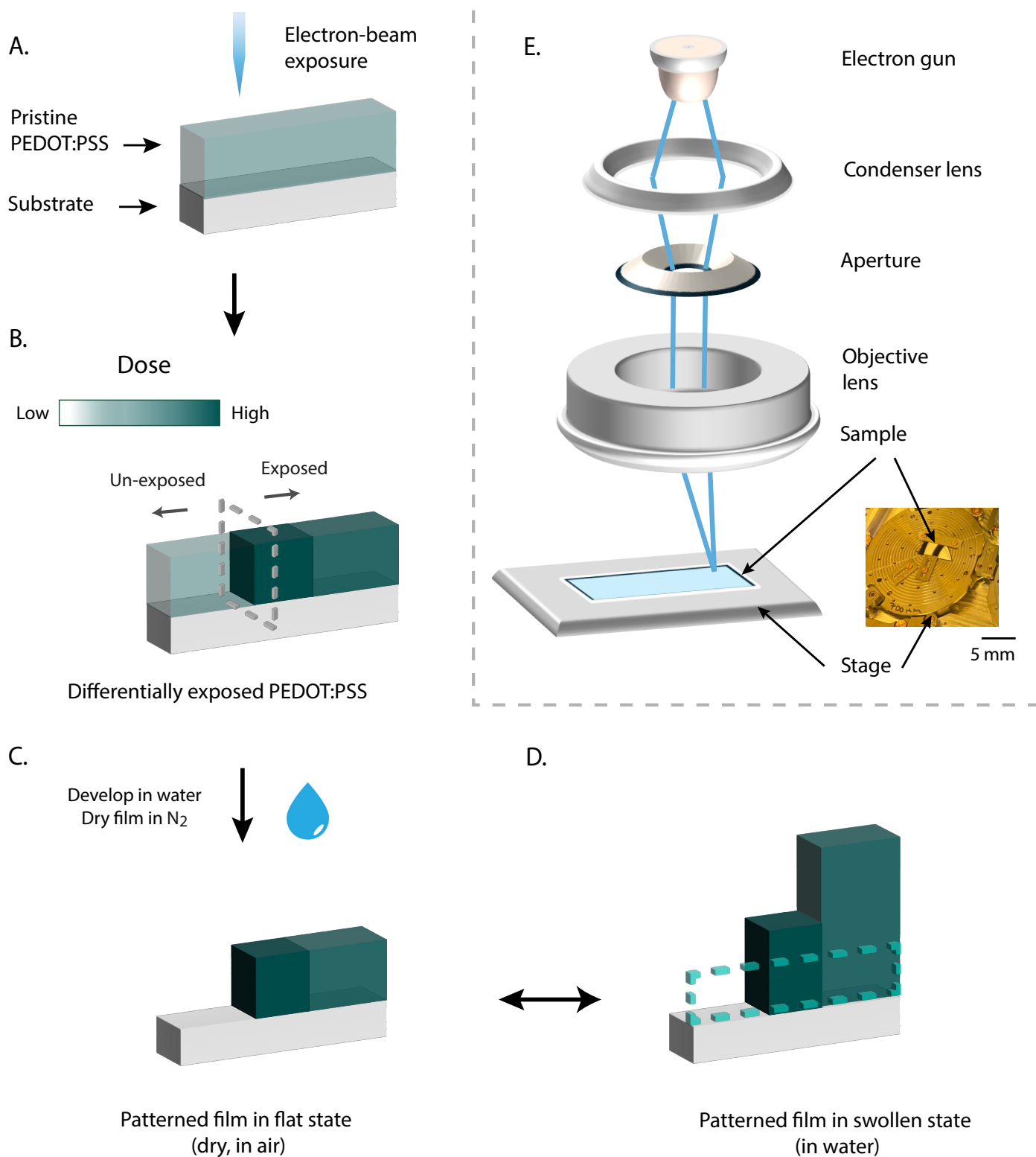
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-025-09948-2>.

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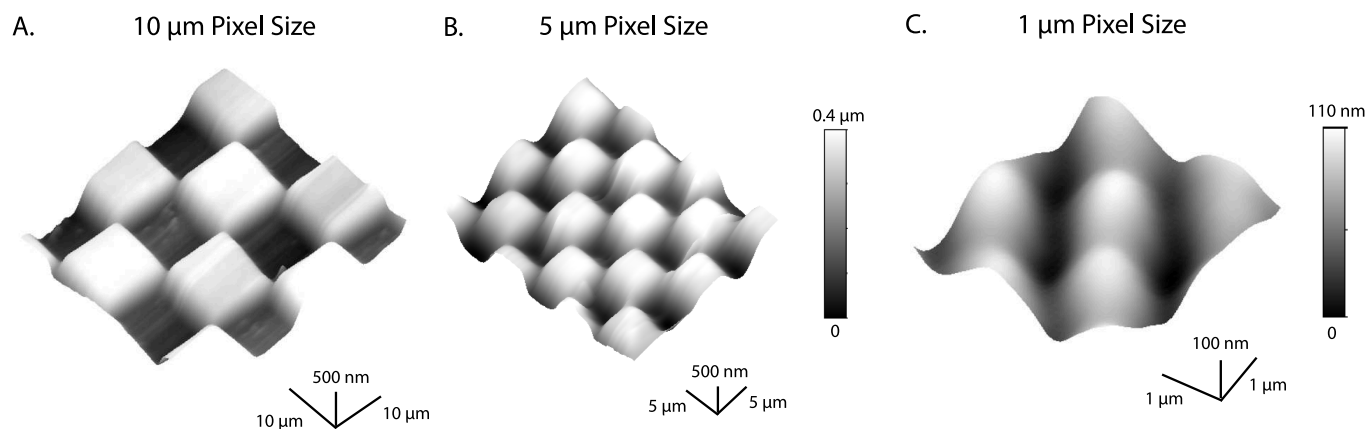
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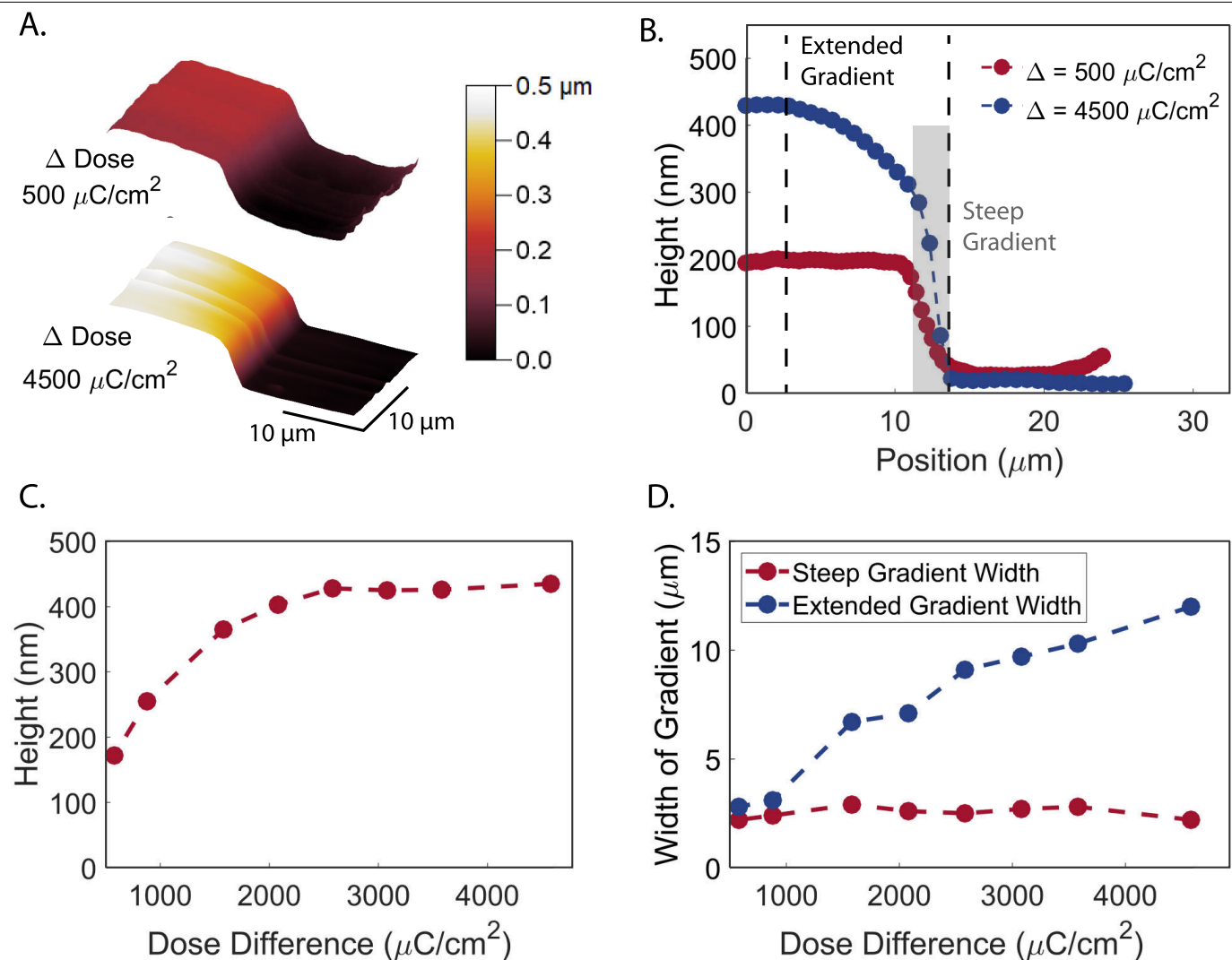
Extended Data Fig. 1 | Electron-beam patterning process flow for programmable surfaces. a–d, Schematic illustrating the fabrication process flow. **a,** Untreated PEDOT:PSS films are dispersible in water. **b,** Exposure to electron-beam radiation stabilizes the material. At or near a critical dose of $700 \mu\text{C cm}^{-2}$, the film is considered fully stabilized, as there is no volume loss

(Supplementary Fig. 1a) after immersion in water. **c,d,** The swelling reduces with increasing dose. **e,** Schematic illustrating the key components of a typical electron-beam lithography tool. The entire system is in a chamber that is held at vacuum. The inset shows the sample secured to the sample holder.



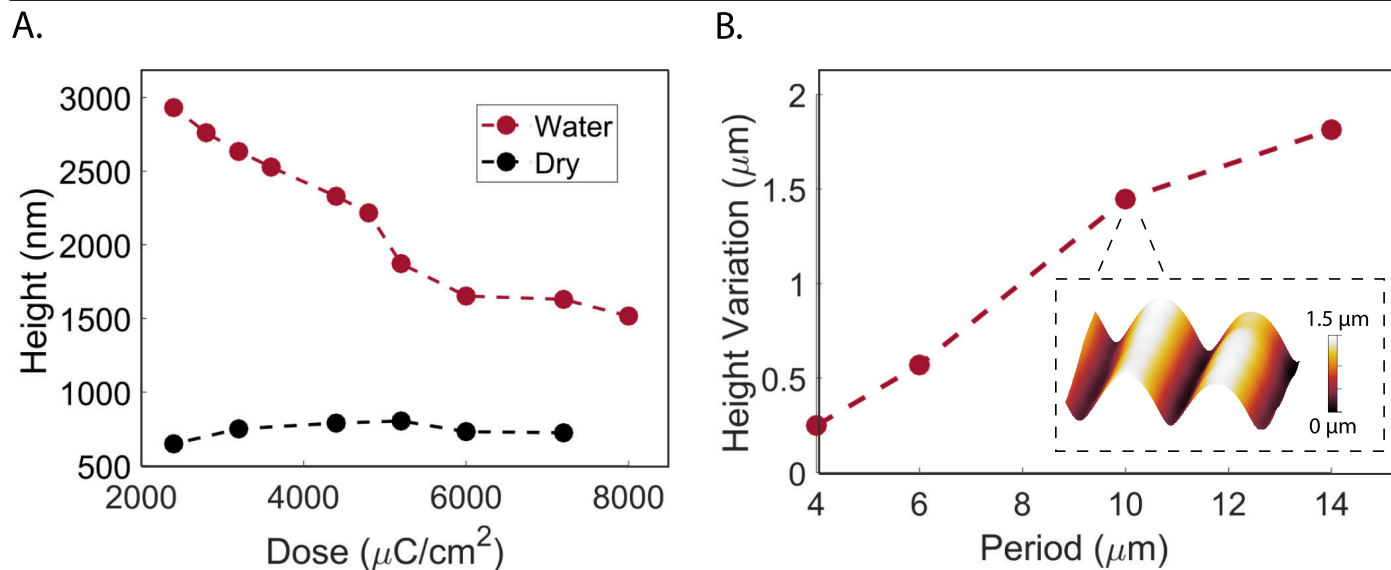
Extended Data Fig. 2 | Checkerboard pattern at different pixel resolutions. AFM images of swollen surfaces encoded with checkerboard patterns of square pixels exposed to alternating doses of 1,000/5,000 $\mu\text{C cm}^{-2}$, with pixel pitches

of 10 μm , 5 μm and 1 μm . Patterns were exposed using a 100-kV acceleration voltage with an initial dry film thickness of roughly 90 nm.



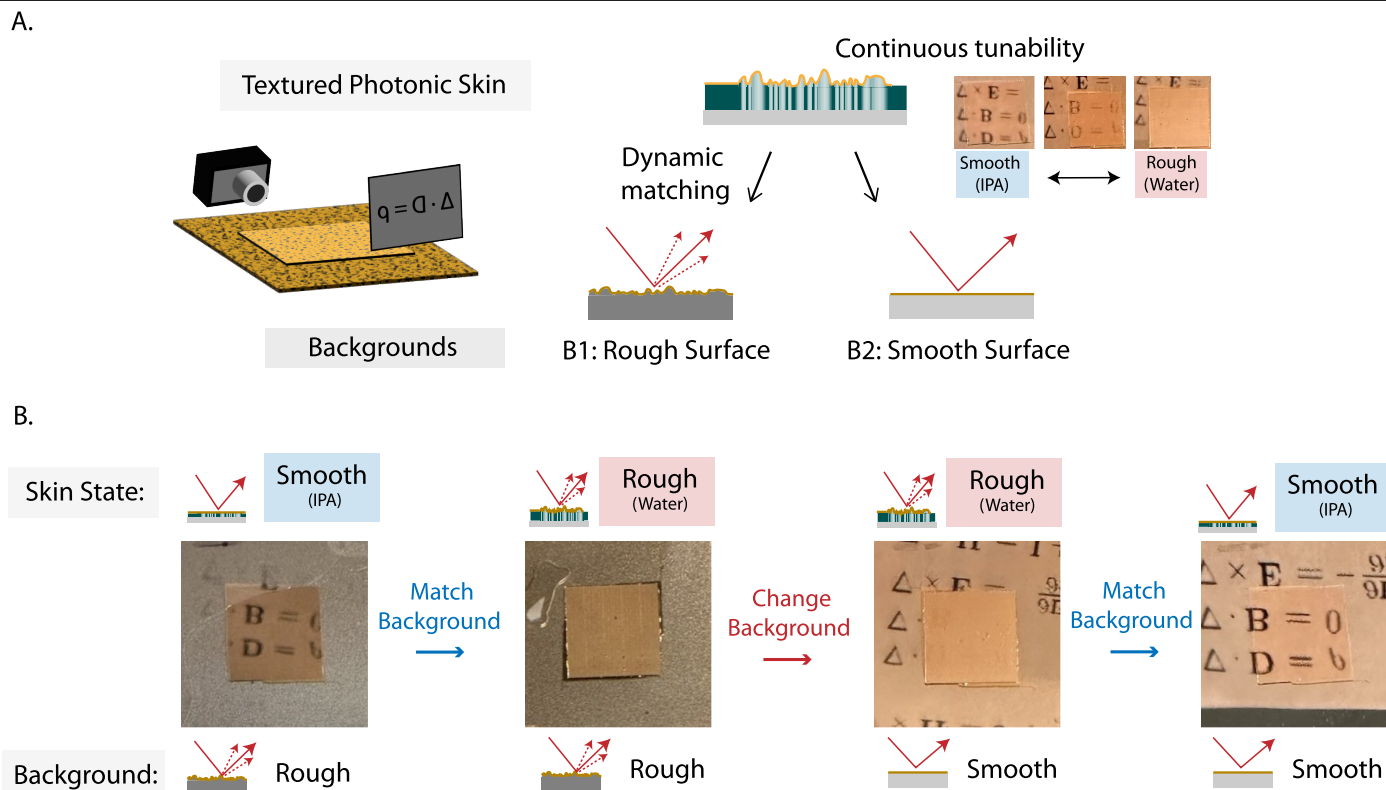
Extended Data Fig. 3 | Dose-dependent height variation and topographic profiles for step interfaces with different applied doses. **a**, In situ liquid AFM measurements of the interface between adjacent rectangles exposed to different dose values. In both examples shown, the lower exposure dose (taller step) was 500 $\mu\text{C}/\text{cm}^2$. The higher dose (shorter step) was exposed to doses of 1,000 $\mu\text{C}/\text{cm}^2$ (upper pattern) and 5,000 $\mu\text{C}/\text{cm}^2$ (lower pattern). Measurements share a common colour bar and scale bar. The exposure was performed using a 50-kV acceleration voltage with an initial dry film thickness

of roughly 90 nm. **b**, Height profiles plotted for the two cases shown in **a**. The dashed black lines show the boundaries of what is considered the 'extended gradient', whereas the grey box shows the boundaries of the 'steep gradient'. **c**, Maximum height difference achieved across the interface as a function of the difference in dose between the higher and lower exposure doses. **d**, Variation of 'steep gradient' and 'extended gradient' width as a function of the difference in dose between the higher and lower exposure doses.



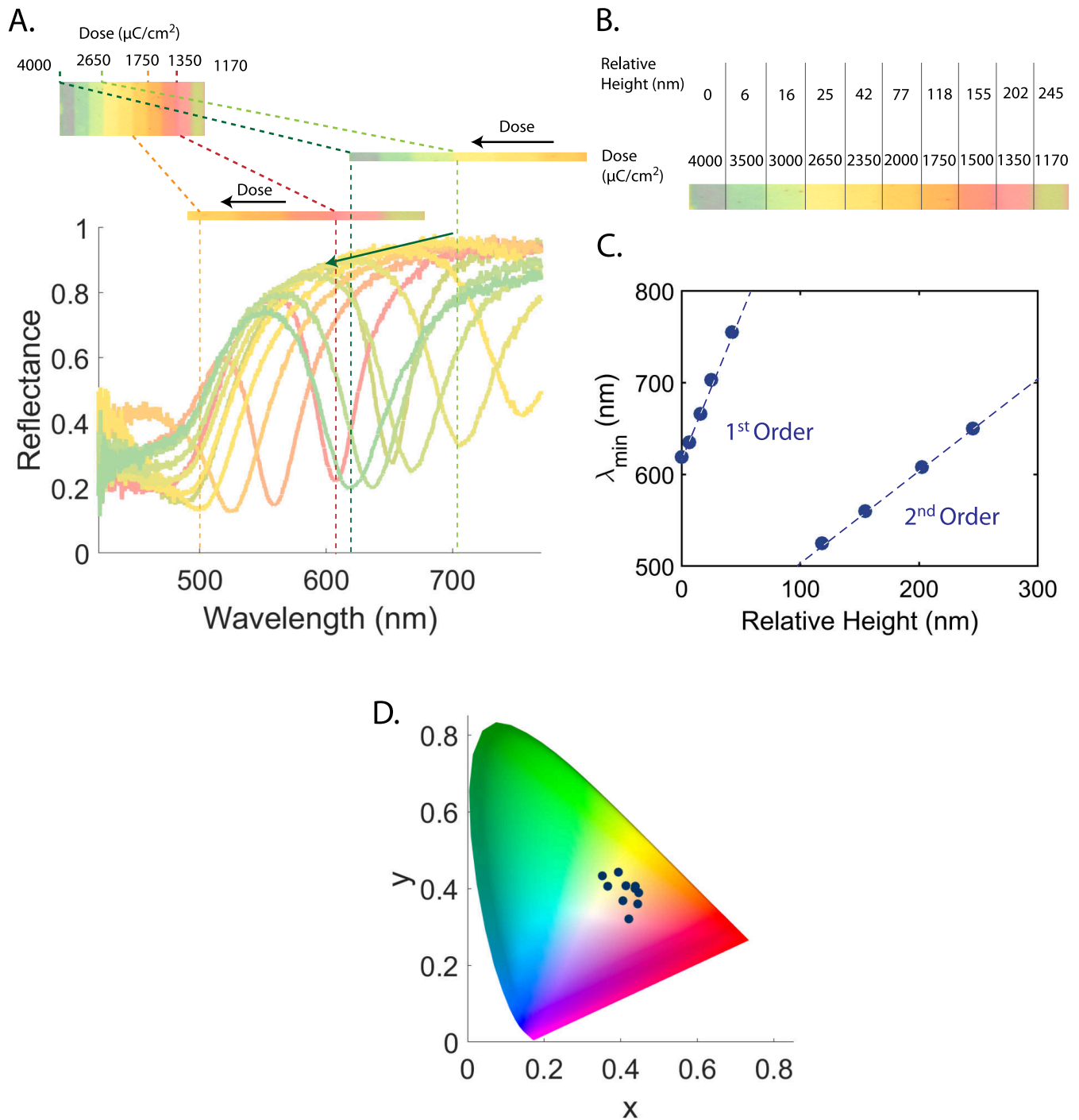
Extended Data Fig. 4 | Patterning of thick films. **a**, Contrast curve for electron-beam-exposed thick PEDOT:PSS with an initial height of about 750 nm. The height of exposed and developed structures in water (red) and in the dry state (black) are plotted as a function of the electron dose. Measurements were made by sampling the edge height of isolated blocks in five locations. Data points

shown are the mean; the standard deviation is not visible as it lies inside the plotted points. Structures were exposed using a 100-kV acceleration voltage. **b**, The achievable height contrast for a square-wave pattern of alternating low-dose ($2,000 \mu\text{C cm}^{-2}$) and high-dose ($10,000 \mu\text{C cm}^{-2}$) stripes at different periods for a film with an initial thickness of around 750 nm.



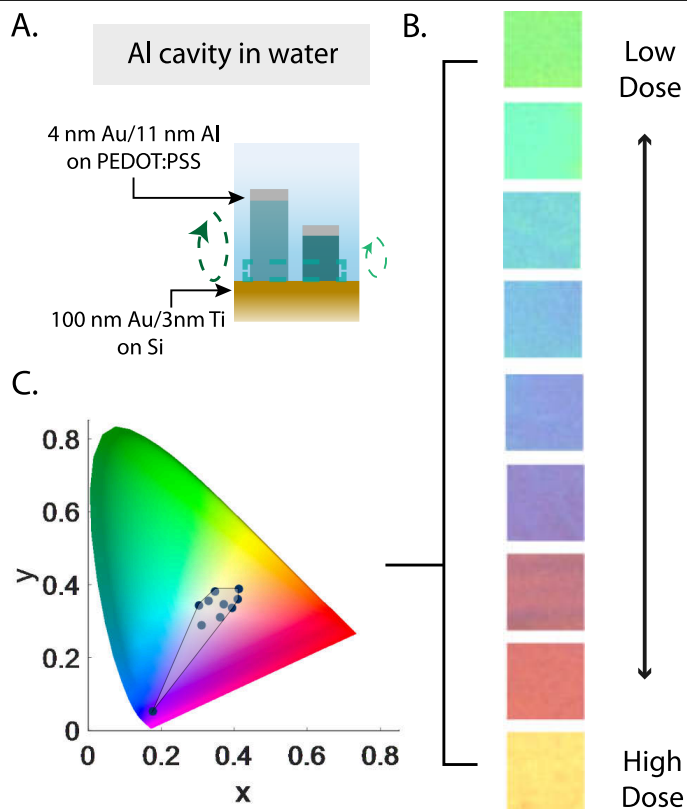
Extended Data Fig. 5 | Dynamic textural background matching between smooth and flat backgrounds. **a**, Schematic of the two different reference backgrounds for smooth and flat states that the dynamic skin aims to match with. The rough side of a Si wafer, coated with 2-nm Ti/20-nm Au, was used as 'B1: rough surface'. The smooth side of a Si wafer, also coated with 2-nm Ti/20-nm Au was used as 'B2: rough surface'. **b**, The photonic skin sample, fabricated as a central square region on a transparent glass coverslip, was

placed on the reference backgrounds. The structure of the photonic skin is described in Fig. 3, consisting of a 20-nm Au film on PEDOT:PSS on glass. Photographs taken with a digital camera of the reflected images formed when printed text of Maxwell's equations are held near the photonic skin in different states on the two reference backgrounds. The smooth background images shown in this figure are the same as in Fig. 3g.

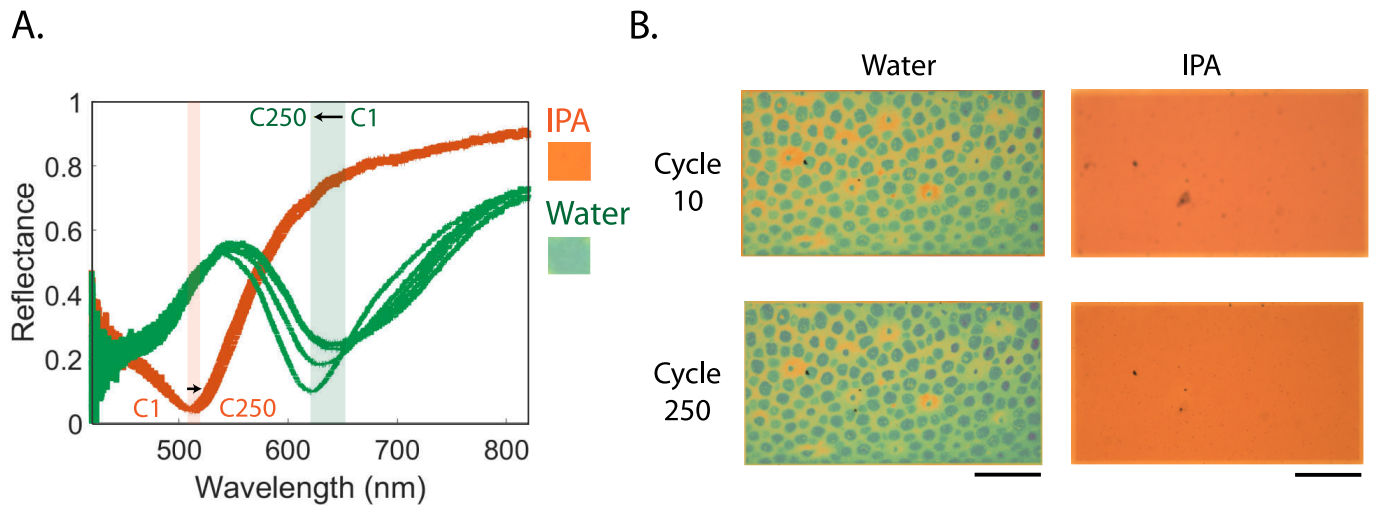


Extended Data Fig. 6 | Modulation of spectral properties and chromaticity of PEDOT:PSS-based optical cavities. **a**, Measured reflectance spectra for electron-beam-patterned PEDOT:PSS-based optical cavities in the swollen state. The colour palette from Fig. 4b is included as an inset, with certain applied doses labelled. The spectra for different stripes of the palette are plotted and linked to their applied dose. Increasing dose, which corresponds to lower swelling, blueshifts the reflectance minima. **b**, Detailed list of doses for strips that show different colours, along with the relative height of the cavities compared with the $4,000 \mu\text{C cm}^{-2}$ strip. The exposure was performed using a 50-kV acceleration voltage with an initial dry film thickness of roughly 90 nm. **c**, The resonance wavelength ($\lambda_{\min}$), determined from the reflectance minimum

of each cavity (**a**), is plotted against relative cavity height (h). Resonances are achieved when the round-trip phase shift in the cavity is an integer multiple of 2π , or $\phi_{\text{total}} = 2\pi m$, in which m is the integer resonance order. The total phase shift combines the propagation phase from a round trip ($\phi_{\text{propagation}} \propto h/\lambda$) and the reflection phase from the mirrors ($\phi_{\text{reflections}}$). When the cavity height is sufficiently large, higher-order resonances begin to appear. The lines of best fit for the data points attributed to the first and second resonant orders yield slopes of 3.28 ($R^2 = 0.999$) and 0.98 ($R^2 = 0.996$), respectively. **d**, The chromaticity coordinates of different strips, determined from their measured spectra, plotted in the CIE 1931 colour space. Many hues, as represented by the angular component of chromaticity, are accessible.



Extended Data Fig. 7 | Modulation of chromaticity of PEDOT:PSS-based optical cavities fabricated using aluminium. **a.** Materials stack of cavities with a hybrid top metal layer of 4-nm Au/11-nm Al. Gold was used as the bottom mirror for the cavity. **b.** Optical micrographs of colours accessible by cavities patterned at applied doses ranging from a lower dose of $1,500 \mu\text{C cm}^{-2}$ to a high dose of $9,000 \mu\text{C cm}^{-2}$. **c.** The chromaticity coordinates of the cavities plotted in the CIE 1931 colour space. A broad range of hues, as represented by the angular component of chromaticity, are accessible. Patterns were exposed using a 100-kV acceleration voltage with an initial dry film thickness of roughly 90 nm.

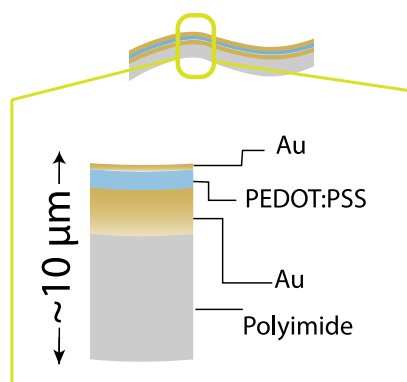


Extended Data Fig. 8 | Switching behaviour over hundreds of cycles.

a. Cavity reflectance spectra sampled over 250 cycles. Measurements were taken at 1, 10, 20, 50, 100 and 250 cycles in water and at 10 and 250 cycles in IPA. Measurements were made on optical cavities with a 20-nm Au top metal layer and a 100-nm Au bottom mirror, on a region with an applied dose of $3,500 \mu\text{C cm}^{-2}$. The orange band indicates the approximately 10 nm spectral shift in the flat state (in IPA) and the green curves show the approximately 25 nm shift in spectral properties in the swollen state (in water). Cavities with this applied

dose of $3,500 \mu\text{C cm}^{-2}$ are expected to show a first-order resonance, in which the dependence of the spectral position on cavity height is $\frac{d\lambda_{\text{min}}(\text{nm})}{dh_{\text{cavity}}(\text{nm})} = 3.28$ (Extended Data Fig. 6). Accordingly, the shift in cavity height is estimated to be about 3 nm in the flat state and about 8 nm in the swollen state. **b.** Image of an optical-cavity-based colour pattern, patterned at doses ranging from 500 to $4,000 \mu\text{C cm}^{-2}$, taken after cycle 10 and cycle 250 in both water and IPA. The exposure was performed using a 50-kV acceleration voltage with an initial dry film thickness of roughly 90 nm. All images use a common scale bar of 100 μm .

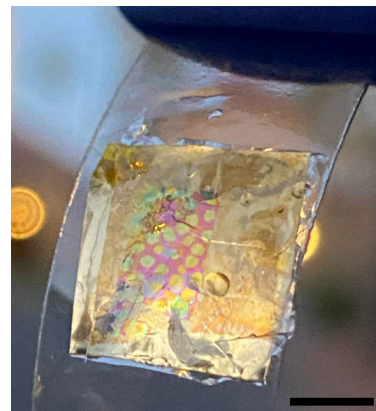
A. Flexible Photonic Skin



B. in IPA



in Water



Extended Data Fig. 9 | Flexible photonic skin. **a**, Schematic of device cross-section for cavity integrated on flexible substrate. **b**, Photographs taken with a digital camera of the device in IPA and water, with a scale bar of 5 mm. A render of the electron-beam pattern shown is included in Supplementary Fig. 10.

Extended Data Table 1 | Summary of exposure dose and key processing conditions for samples shown in the figures in the main text

<u>Figure</u>	<u>Acceleration Voltage</u>	<u>Dry Film Thickness</u>	<u>Dose Range</u>
2C	50 kV	90 nm	250 $\mu\text{C}/\text{cm}^2$ to 3500 $\mu\text{C}/\text{cm}^2$ (Fig. S4)
2D	50 kV	90 nm	2500 $\mu\text{C}/\text{cm}^2$ / 500 $\mu\text{C}/\text{cm}^2$
2E	50 kV	90 nm	4000 $\mu\text{C}/\text{cm}^2$ (central tree), 1000 $\mu\text{C}/\text{cm}^2$ (letter “S”), 2500 $\mu\text{C}/\text{cm}^2$ (planar background)
3	50 kV	90 nm	250 $\mu\text{C}/\text{cm}^2$ to 3500 $\mu\text{C}/\text{cm}^2$, per-pixel dose assigned using BEAMER software
4A-B	50 kV	90 nm	From right to left, from 1175 $\mu\text{C}/\text{cm}^2$ to 4000 $\mu\text{C}/\text{cm}^2$, in dose steps that begin at 100 $\mu\text{C}/\text{cm}^2$ and increase geometrically by 25%.
4D	100 kV	90 nm	7500 / 4750 $\mu\text{C}/\text{cm}^2$ for the “Small Dose Δ ” 7500 / 1500 $\mu\text{C}/\text{cm}^2$ for “High Dose Δ ”
4G-I	50 kV	90 nm	250 $\mu\text{C}/\text{cm}^2$ to 3500 $\mu\text{C}/\text{cm}^2$, per-pixel dose assigned using BEAMER software
4L-M	100 kV	120 nm	1000 $\mu\text{C}/\text{cm}^2$ to 7000 $\mu\text{C}/\text{cm}^2$, per-pixel dose assigned using BEAMER software
5	50 kV	90 nm	250 $\mu\text{C}/\text{cm}^2$ to 3500 $\mu\text{C}/\text{cm}^2$